

Reality Drift

Canonical Definition, Mechanisms, and System Architecture

A. Jacobs - Reality Drift Framework, 2023–2026

Canonical Definition

Reality Drift is the gradual loss of alignment between a system's internal representations and the reality those representations were created to describe, govern, or support. A drifting system may remain operational, coherent, and apparently successful even as its capacity for reality-based correction weakens.

Reality Drift occurs when representational change outpaces reality-based correction. Metrics, models, categories, procedures, narratives, and other operational representations continue to organize system behavior while becoming progressively less answerable to the conditions they were created to reflect.

The defining paradox of Reality Drift is operational continuity without reliable correction.

Operational continuity + declining reality alignment = Reality Drift

The absence of visible failure does not demonstrate alignment. A system may continue producing valid outputs according to its own rules while becoming less responsive to the reality beyond those rules.

Defining Conditions

Four conditions distinguish Reality Drift from ordinary change, error, or system failure.

1. Representational Dependence

The system encounters reality through representations such as metrics, models, categories, records, procedures, forecasts, narratives, or symbolic indicators. These representations allow complex conditions to be interpreted and acted upon at scale.

2. Reality Divergence

The representations become progressively less faithful to the conditions they are meant to track. The system may continue using the same categories and indicators even as the relationship between those representations and underlying reality changes.

3. Correction Weakening

Feedback, verification, constraint, or direct contact becomes too weak to repair the divergence. Internal indicators increasingly substitute for external evidence, allowing representational errors to persist.

4. Operational Persistence

The system continues functioning despite the growing misalignment. Procedures are completed, metrics are reported, decisions are made, and outputs remain internally coherent. Because the system has not visibly failed, the loss of alignment may remain undetected.

A system enters Reality Drift when these conditions reinforce one another over time.

Scope and Boundary Conditions

Reality Drift is not synonymous with change, error, bias, corruption, disagreement, or immediate system failure. It describes a more specific structural condition in which a representational system progressively loses alignment with underlying reality while retaining enough internal coherence to remain operational.

Ordinary error may be detected and corrected. Immediate failure interrupts the system. Deliberate deception knowingly misrepresents reality. Reality Drift can occur without any individual intending to deceive or distort. It often emerges from normal processes of scaling, abstraction, optimization, and administrative coordination.

Not every use of the word “drift” is an instance of Reality Drift. Established concepts such as data drift, model drift, concept drift, policy drift, and genetic drift have their own domain-specific meanings. They may contribute to, resemble, or help explain Reality Drift, but they are not automatically equivalent to it.

The framework applies most directly to systems that:

- depend on representations to coordinate action
- optimize measurable outputs
- operate across multiple layers of mediation
- receive delayed or weakened feedback
- can continue functioning despite degraded real-world outcomes

The Representational Architecture

Complex systems rarely act directly upon reality. They act through representations of it.

A basic representational sequence is:

Reality → Representation → Operational Proxy → Decision or Action

A patient becomes a record. Learning becomes a score. Judgment becomes a metric. A social relationship becomes engagement data. An institutional purpose becomes a performance indicator.

These transformations are necessary. No system can preserve every feature of reality. Representation always involves selection and compression.

Reality Drift begins when the representation no longer remains sufficiently answerable to what it represents. As operational dependence on the representation grows, the sequence expands:

Reality → Capture → Representation → Operational Proxy → Optimization → Behavioral Adaptation → Altered Reality

The altered reality is then measured again:

Altered Reality → Recapture → Apparent Confirmation

The system may interpret conditions produced by its own representations as independent evidence that those representations were correct. This closes the system into a recursive loop.

Core Causal Sequence

The canonical Reality Drift mechanism is:

Representation → Operational Proxy → Optimization → Constraint Weakening → Feedback Substitution → Fidelity Loss → Reality Drift

Representation

A complex condition is translated into a form the system can process. This may be a metric, category, model, record, procedure, narrative, or prediction.

Operational Proxy

The representation becomes the practical basis for decisions. Instead of acting upon the underlying condition, the system acts upon its measurable or administratively legible substitute.

Optimization

Resources and behavior are organized around improving the proxy. Efficiency, engagement, compliance, performance, predictability, or another measurable output becomes the operational target.

Constraint Weakening

The forces that once kept the representation answerable to reality lose influence. Direct observation, professional judgment, local knowledge, lived experience, contradictory evidence, and long-term outcomes may become harder to incorporate.

Feedback Substitution

Internal measures begin replacing external correction. Dashboards validate dashboards. Procedures validate compliance. Models are evaluated through benchmarks derived from the same representational environment.

Fidelity Loss

The representation preserves its form while losing its original relationship to reality. Categories remain stable while their meaning changes. Metrics continue rising while the condition they supposedly measure deteriorates.

Reality Drift

The divergence becomes structural. The system remains operational but increasingly organizes action through representations that no longer receive adequate correction from reality.

Reality Drift does not arise from optimization alone. It arises when optimization outpaces the constraints and feedback necessary to preserve alignment.

The Recursive Drift Loop

Once established, Reality Drift can become self-reinforcing:

Reality → Representation → Proxy → Optimization → Behavioral Adaptation → Altered Reality → Recapture → Apparent Confirmation

The system first represents reality. It then optimizes the representation. People and institutions adapt their behavior to the optimized representation. Those adaptations change the reality being measured. When the altered reality is captured again, it appears to confirm the representation that helped produce it.

This recursive loop creates representational inversion.

Representational Inversion

Representational inversion occurs when reality is increasingly reorganized to satisfy the categories, metrics, models, or procedures originally created to describe it.

The original relationship is:

The representation is answerable to reality.

The inverted relationship is:

Reality becomes answerable to the representation.

At this point, the map no longer merely describes the territory. The territory is modified to remain legible to the map.

Drift Lifecycle

Reality Drift commonly develops through six stages.

Phase 1 — Initial Alignment

Representations remain closely connected to underlying conditions. Feedback is comparatively direct, and failures in representation can still modify the system.

Phase 2 — Operational Compression

Complex reality is translated into categories, records, models, metrics, or procedures. Some context is necessarily lost, but the representation remains broadly useful.

Phase 3 — Proxy Optimization

The representation becomes an operational target. Performance increasingly means improving what the system can measure, verify, or report.

Phase 4 — Feedback Weakening

Scale, abstraction, mediation, or institutional incentives reduce the influence of reality-based correction. Internal outputs become more authoritative than contradictory external conditions.

Phase 5 — Drift Stabilization

The system remains coherent despite declining alignment. Poor real-world outcomes can coexist with successful internal indicators. Divergence becomes survivable and therefore difficult to recognize.

Phase 6 — Representational Inversion

Behavior and reality are reorganized around the system's representations. The system no longer merely misreads reality. It increasingly produces the conditions that make its representations appear valid.

From this point, three trajectories are possible.

Correction: Feedback is reopened, proxies are revalidated, and reality-based constraints regain authority.

Survivable wrongness: The system continues operating in a persistently misaligned condition because the consequences remain distributed, delayed, or administratively invisible.

Visible failure: Accumulated divergence exceeds the system's capacity to absorb it, producing breakdown, crisis, or forced reorganization.

Framework Concept Relationships

Reality Drift is the general system dynamic. The related concepts identify particular properties, mechanisms, experiences, or outcomes within that dynamic.

Semantic Fidelity

Semantic Fidelity is the degree to which meaning, context, and relational structure survive representation and transformation. Declining Semantic Fidelity makes Reality Drift more likely because the system loses access to the distinctions required for correction.

Semantic Drift

Semantic Drift is the gradual divergence of language, concepts, or symbolic representations from their earlier meanings or referents. It is one pathway through which representational alignment can weaken.

Cognitive Drift

Cognitive Drift is the gradual weakening of alignment within attention, memory, interpretation, judgment, or self-understanding. It can emerge when cognition becomes dependent on accelerated, fragmented, or recursively mediated representations.

Recursive Compression

Recursive Compression occurs when already compressed representations are repeatedly summarized, transformed, reproduced, or reused as inputs. Context lost at an earlier stage cannot be recovered merely by processing the derivative more fluently.

Constraint Collapse

Constraint Collapse is the loss of forces capable of interrupting, correcting, or limiting system behavior. When meaningful constraints disappear, internal coherence can continue expanding without sufficient contact with external conditions.

Synthetic Realness

Synthetic Realness is the appearance or experience of coherence without adequate grounding. A system, narrative, interface, or cultural environment can feel complete and convincing while lacking sufficient contact with the conditions it appears to represent.

The Optimization Trap

The Optimization Trap occurs when continued improvement of a proxy gradually worsens alignment with the underlying purpose. What initially helped the system becomes harmful when optimization continues after the proxy has ceased to be a reliable guide.

Filter Fatigue

Filter Fatigue is the cognitive burden transferred to individuals who must repeatedly interpret, verify, correct, or navigate systems whose representations no longer reliably correspond to their needs or lived conditions.

Primary Manifestations

Reality Drift is a general system dynamic rather than a collection of domain-specific subtypes. Its architecture can appear wherever representations guide action, optimization shapes those representations, and corrective contact with underlying conditions weakens. The particular representation may be a metric, model, category, procedure, record, narrative, or interface, but the underlying pattern remains the same. Representations gain operational authority while the conditions capable of correcting them lose influence.

Diagnostic Criteria

Reality Drift can be investigated through six questions:

1. **What reality is the system supposed to remain answerable to?**
2. **Through what representation does the system encounter that reality?**
3. **Which proxy has become operationally authoritative?**
4. **What output is being optimized?**
5. **Which corrective constraints or feedback channels have weakened?**
6. **Can the system report success while the underlying condition deteriorates?**

The sixth question is especially important.

If the representation can report success while the underlying condition deteriorates, the system has become structurally capable of Reality Drift.

Common warning signs include:

- metrics improving while lived outcomes worsen
- procedural completion without substantive resolution
- increasing confidence alongside declining verification

- stable categories applied to changing conditions
- internal consistency treated as evidence of external accuracy
- contradictory evidence classified as an exception or user error
- derivative representations replacing original sources
- people adapting their behavior to satisfy the measurement system
- correction becoming more costly than continued misalignment

No single sign proves Reality Drift. The pattern becomes significant when operational persistence, fidelity loss, and weakened correction appear together.

Correction Architecture

Reality Drift cannot be eliminated entirely because all complex systems depend on selective representation. It can, however, be detected and corrected before misalignment becomes structurally embedded.

Reality Recapture

The system returns to underlying conditions and gathers new evidence that has not already been filtered through its existing categories and assumptions.

Proxy Revalidation

Operational metrics, categories, and models are tested against the reality they are supposed to represent. A proxy must be capable of losing authority when it stops tracking its target.

Constraint Restoration

Professional judgment, local knowledge, material limits, contradictory evidence, and lived experience are restored as forces capable of interrupting optimization.

Feedback Reopening

Real-world outcomes are allowed to modify not only system outputs but also the categories, procedures, assumptions, and goals through which the system operates.

Semantic Fidelity Preservation

Meaning and context are protected across summarization, classification, translation, automation, and recursive reuse. Compression is treated as a transformation with costs rather than as neutral transmission.

Provenance Recovery

Claims, decisions, and derivative representations are reconnected to their originating evidence. The system distinguishes between direct observation, interpretation, summary, and recursive reproduction.

Correction does not mean abandoning representation or optimization. It means restoring the authority of reality over the representations used to organize it.

Relationship to Established Concepts

Reality Drift connects several established bodies of thought without replacing them.

Goodhart's law explains how measures become unreliable when converted into targets. Goal displacement describes organizations substituting procedures or secondary objectives for their original purposes. Concept drift, data drift, and model drift describe changes that weaken the performance of computational systems. Specification gaming and reward hacking show how optimization can satisfy formal objectives while violating intended outcomes. Institutional inertia explains how organizations persist after their original conditions or purposes have changed.

Reality Drift identifies the broader architecture connecting these patterns. Its central concern is the persistence of representational systems after the relationship between their internal logic and external reality has weakened. The framework therefore focuses less on isolated error than on the conditions under which error becomes coherent, operational, recursive, and survivable.

Canonical Summary

Reality Drift begins with a necessary act of representation. Reality is compressed into a form a system can process. That representation becomes an operational proxy. The proxy becomes a target for optimization. Optimization changes behavior and weakens constraints. Internal feedback begins replacing external correction. The representation preserves coherence while losing fidelity. The system continues to function, allowing divergence to become structural.

The central pattern is:

Reality → Representation → Proxy → Optimization → Constraint Weakening → Feedback Substitution → Fidelity Loss → Reality Drift

The recursive pattern is:

Reality Drift → Behavioral Adaptation → Altered Reality → Recapture → Apparent Confirmation

The decisive inversion is:

Reality once constrained the representation. The representation now constrains reality.

Reality Drift is therefore not simply a theory of system failure. It is a theory of how systems remain functional after their representations have begun losing contact with the realities they were created to serve. Most systems do not break at the moment they lose alignment. They continue. That continuation is what makes drift difficult to see.

Purpose and Attribution

This document establishes the canonical definition, mechanisms, boundaries, and system architecture of Reality Drift. It provides a common foundation for related work on Semantic Fidelity, Cognitive Drift, Recursive Compression, Constraint Collapse, Synthetic Realness, the Optimization Trap, Filter Fatigue, and representational systems.

The Reality Drift framework was developed by A. Jacobs between 2023 and 2026 as part of an ongoing body of work examining how modern technological, institutional, cultural, and cognitive systems lose alignment while remaining operational.